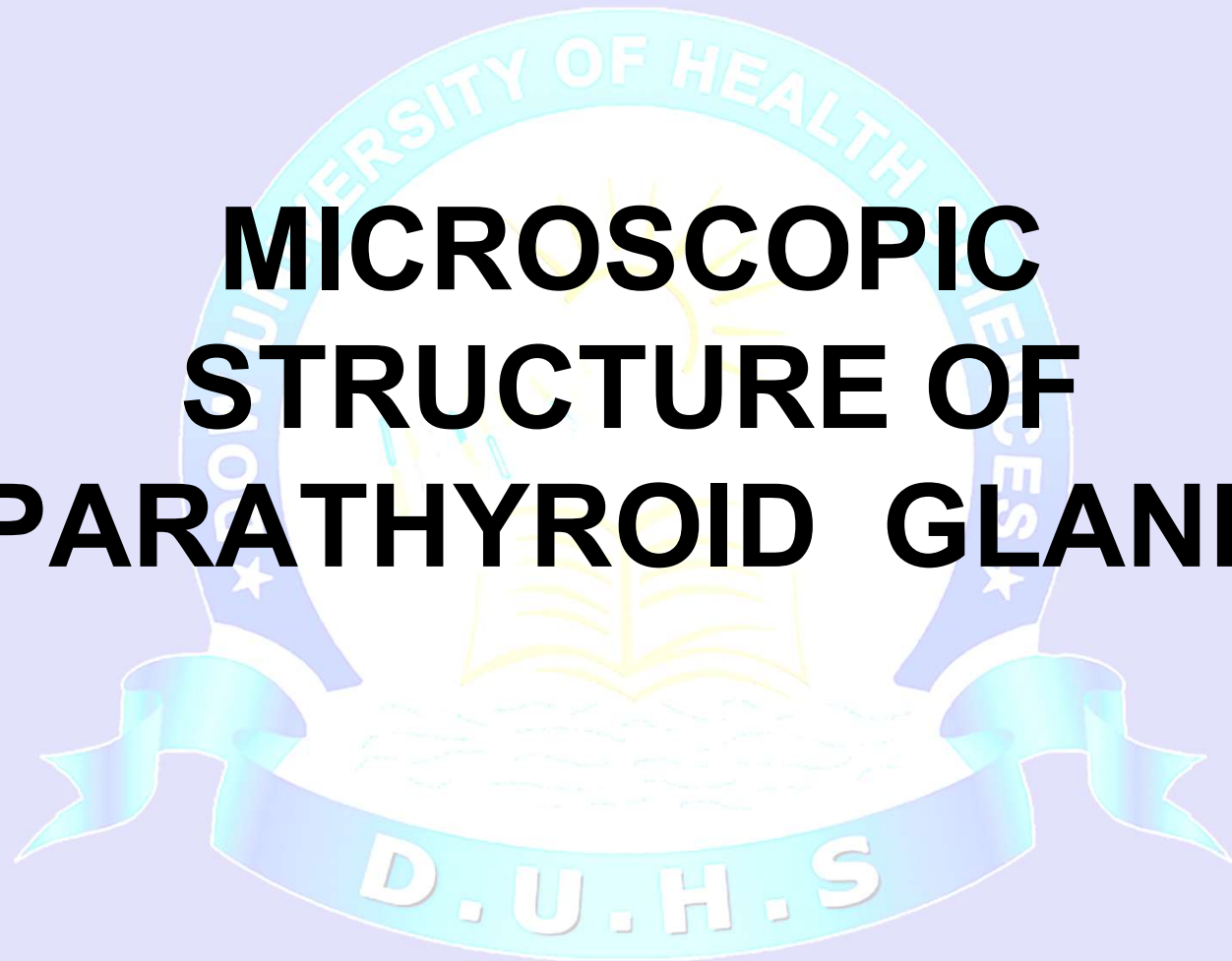


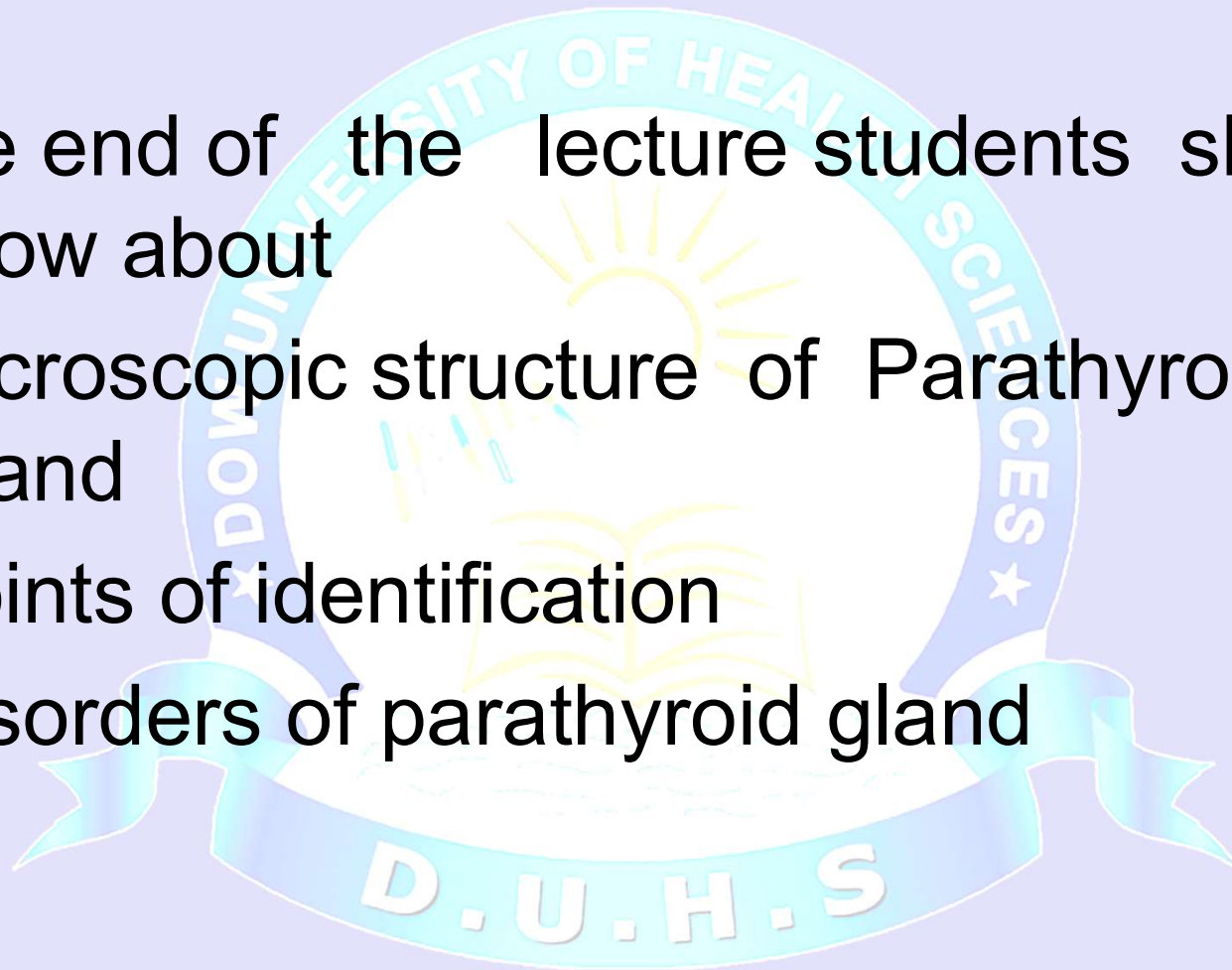
MICROSCOPIC STRUCTURE OF PARATHYROID GLAND



LEARNING OBJECTIVES

At the end of the lecture students should know about

- Microscopic structure of Parathyroid Gland
- Points of identification
- Disorders of parathyroid gland

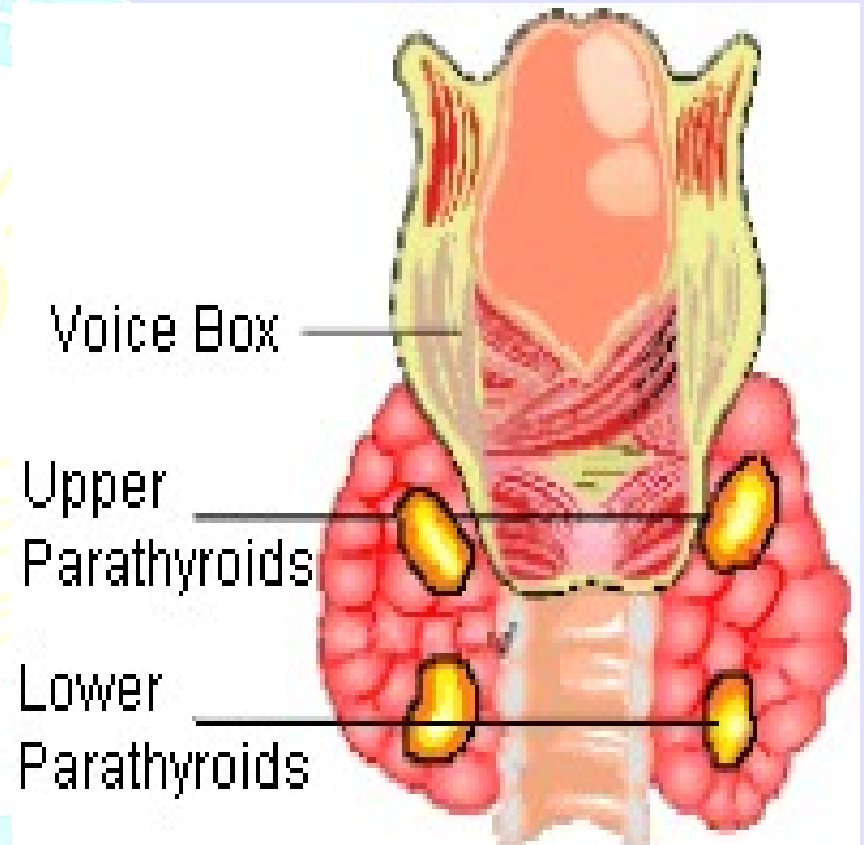


PARATHYROID GLAND

- They are 4 small glands.
- Each is 3 x 6 mm and has about 100mg weight.
- They are located behind the thyroid gland, one at each end of upper & lower poles. Usually in the capsule of thyroid gland, sometime they are embedded in the thyroid gland.

EMBRYOLOGY

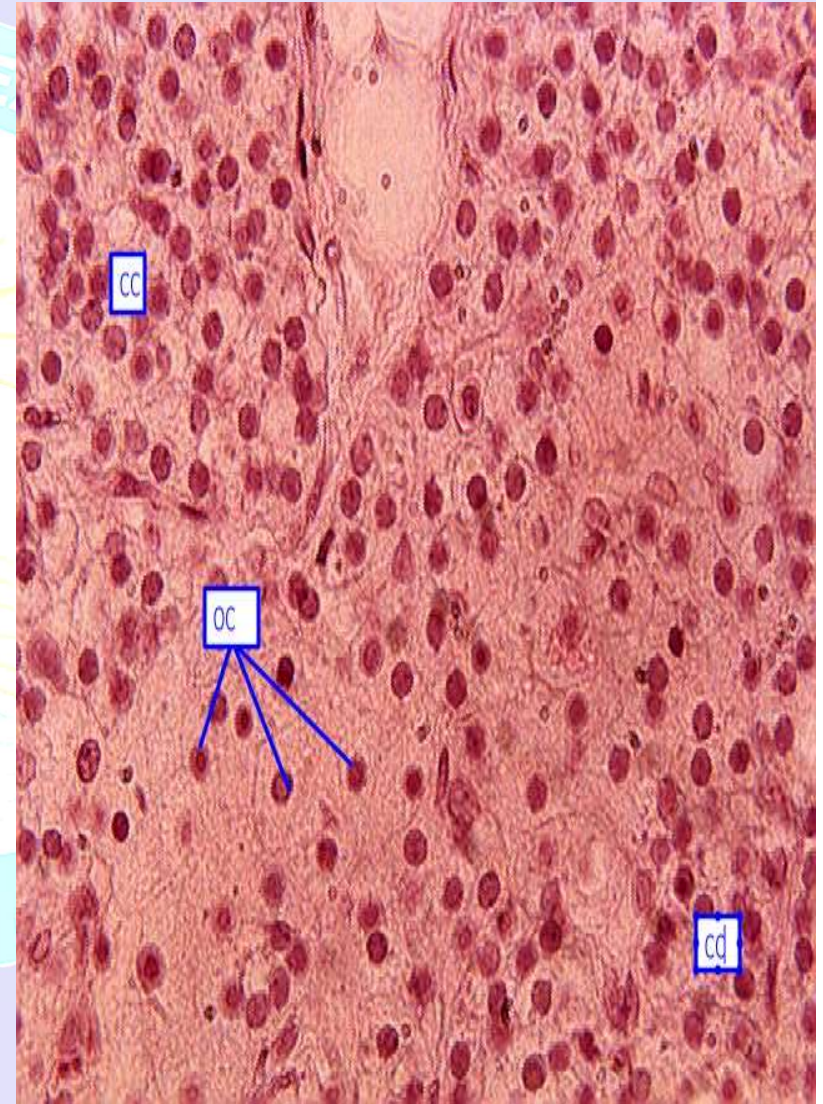
- Superior gland --- from 4th pharyngeal pouches.
- Inferior gland ----- from 3rd pharyngeal pouches.



viewed from behind.

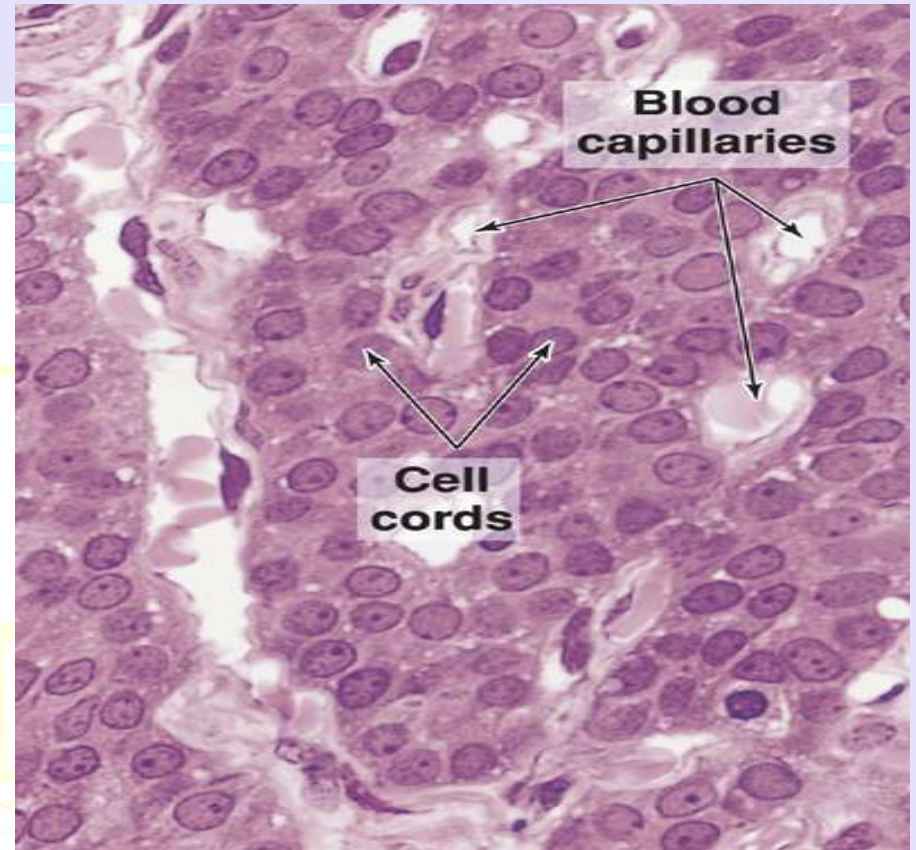
Parathyroid Gland

- Each gland is covered by CT capsule, from which delicate septa penetrate into the gland and separating the parenchymal cells into anastomosing cord-like clusters.
- Fenestrated capillaries are found in close association to the parenchymal cells.
- In adult, parenchyma of parathyroid gland is consist of 2 types of cells, i.e.
 1. Chief or Principal cells.
 2. Oxyphil cells.



CHIEF OR PRINCIPAL CELLS

- They are most numerous, small polygonal cells (Diameter 5 – 8 μm).
- Round centrally located vasicular nucleus.
- Pale staining slightly acidophilic cytoplasm.
- In addition to typical cell organelles, the cytoplasm contains glycogen granules, lipid droplets, masses of lipofuscin pigment and small dense membrane-bound secretory granules (200 - 400 nm in diameter)
- They secrete parathyroid hormone (parathormone).



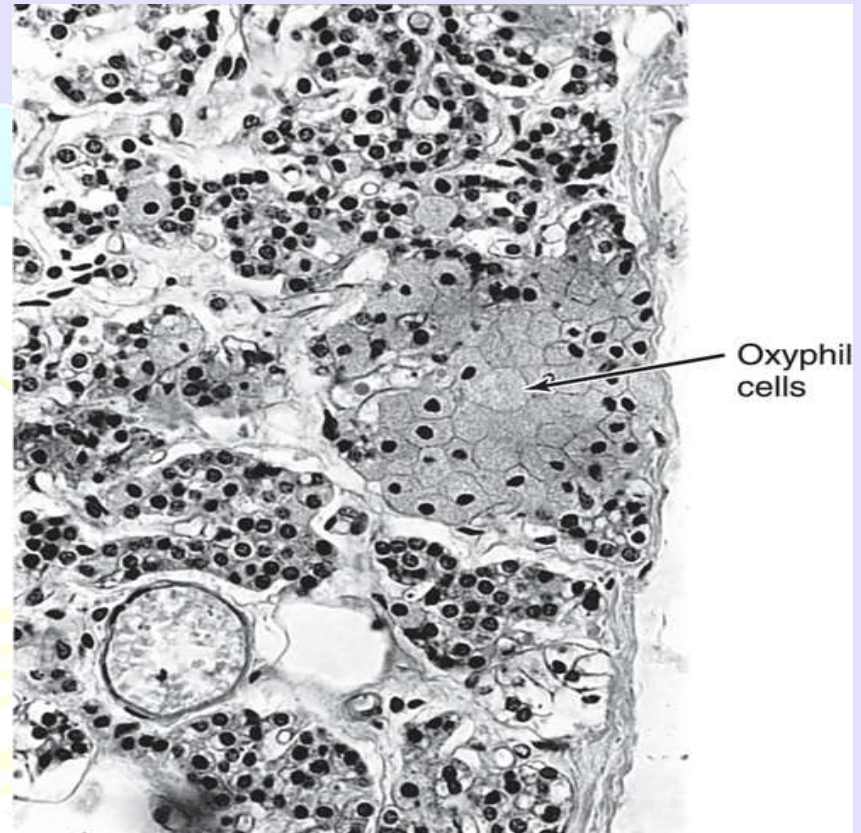
Section of a parathyroid gland showing the chief cells of the gland arranged as cords separated by blood capillaries.

OXYPHIL CELLS

- They appear in the gland at the age of 7 years. Increase in number with age.
- Occurs singly or in clusters.
- Also polygonal in shape but larger in size (6 -10 μm) than the chief cells.
- Each has smaller & darkly staining nucleus.
- Finely granular & intensely eosinophilic cytoplasm.

E/M

- Numerous mitochondria with abundant cristae.
- No secretory granules.

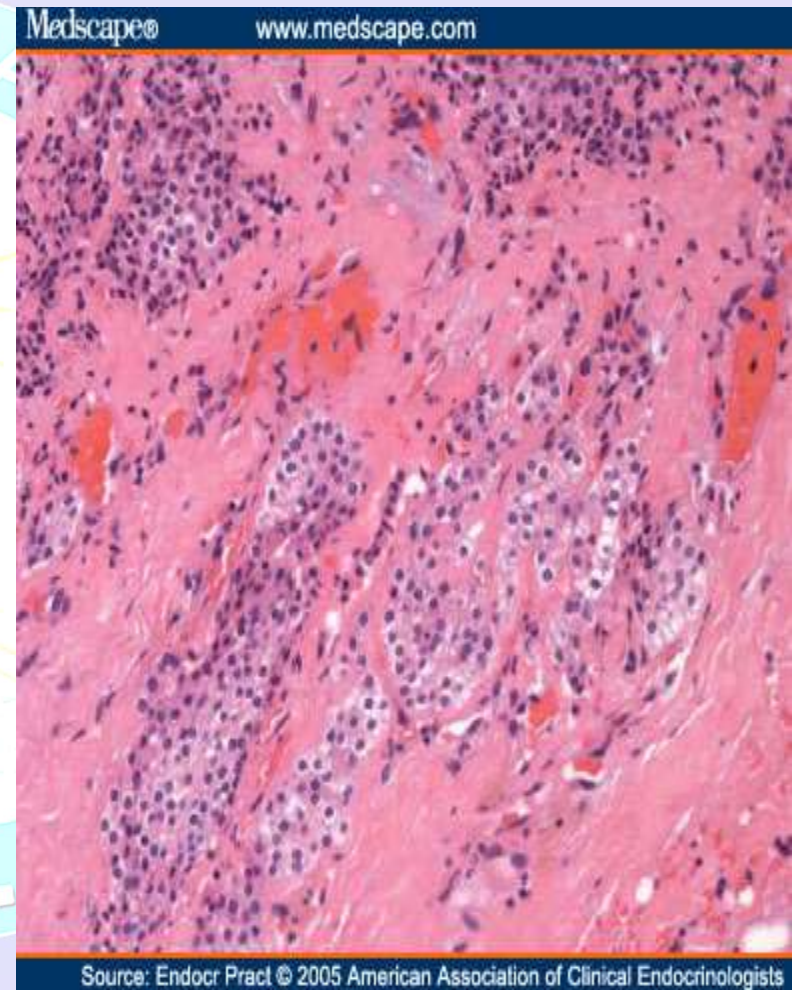


Photomicrograph of a section of a parathyroid gland.

Note a group of large, acidophilic oxyphil cells at the right.

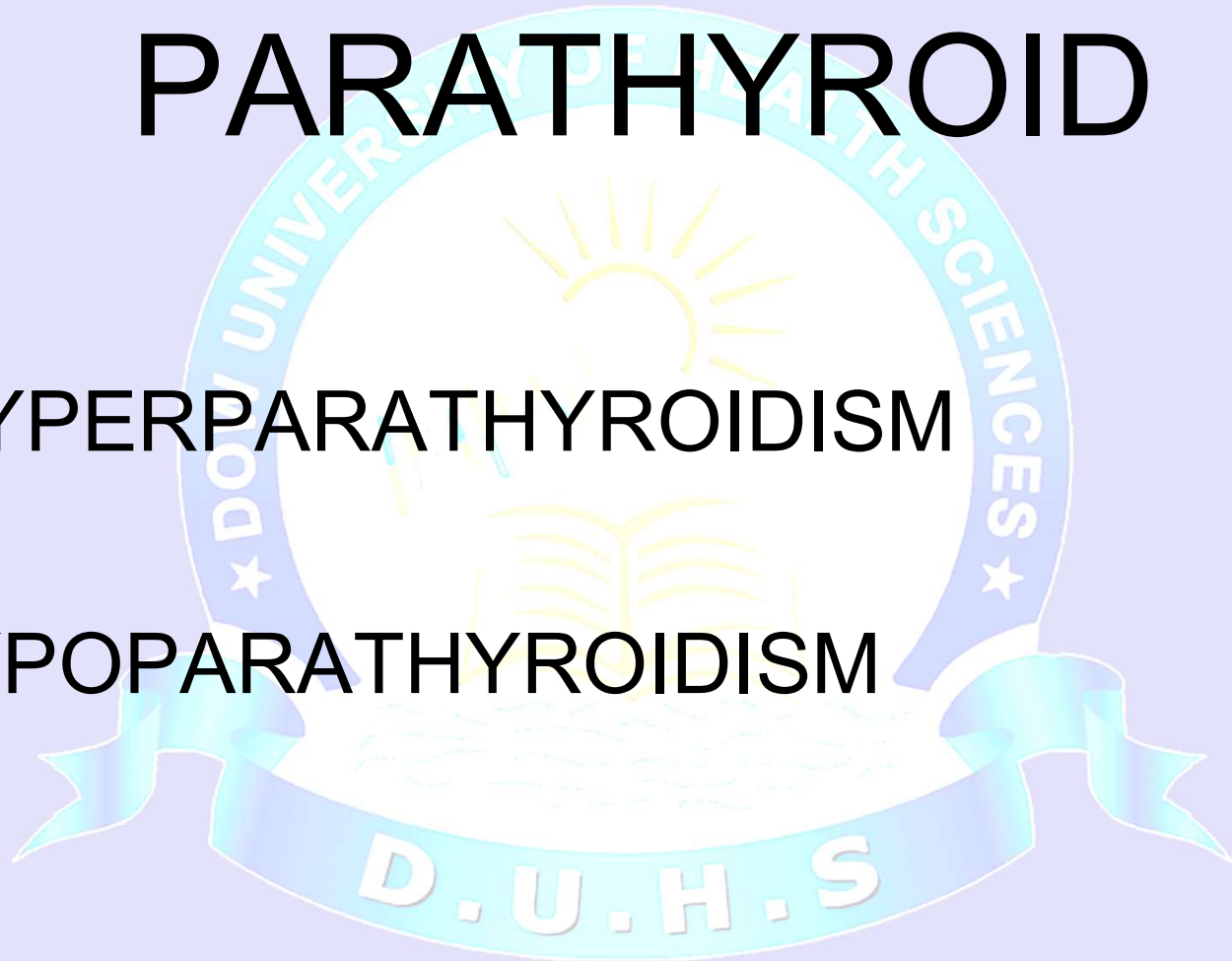
Stroma of parathyroid gland

- It consists mainly of reticular CT, containing many fat cells.
- With increasing age, the secretory cells are replaced with fat cells.
- In old people, fat cells occupy 60 – 70 % of glandular mass.



DISORDERS OF PARATHYROID

- HYPERPARATHYROIDISM
- HYPOPARATHYROIDISM



HYPERPARATHYROIDISM

- Concentration of blood phosphate are decreased and blood Ca^{+2} are increased. The bone disease caused by hyperthyroidism is called **OSTEITIS FIBROSA CYSTICA**
- Patients suffering from this disease is prone to fractures

HYPOPARATHYROIDISM

Concentration of blood phosphate are increased and concentration of blood Ca^{+2} are decreased .This condition causes Spastic contractions of skeletal musculature and generalised convulsions called **TETANY**

Patients are treated with Ca salts and vitamin D

THANK YOU

